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Marketing A/B Testing Analysis using Python

For this task, I used a **marketing dataset** that contains information about users exposed to two campaigns, **ad (control group)** and **psa (test group)**.

The main objective was to **compare the performance** of both campaigns using statistical analysis.

First, I loaded the dataset and checked its structure. Then, I performed data cleaning by converting the converted column into numeric format and removing missing values.

Then, I created a summary table to get total users and conversions for each group. **Converted True / False to 1/0.**

Next, I calculated the **conversion rate** for each group using the **mean function**. I observed that the **ad group** has a **higher conversion** rate than the **psa group**. I also plotted a **bar chart and line chart** to compare the conversion rates between the two groups.

After that, I performed a two-proportion **z-test** to check statistical significance. The **p-value was very small**, showing a difference

I also calculated lift, which showed that the ad campaign performs better. The confidence interval did not include zero, which confirms the result.

Then, I performed a chi-square test, which supported the same conclusion.

I also checked **SRM**, but since the dataset does **not provide an expected split**, the **SRM test was not applicable**.

The **ad campaign performs better than the psa campaign**, as per results by statistical tests and visualizations. Therefore, **the ad campaign should be continued**.